ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B.Tech. (CSE-AI)

Semester: IV

SUBJECT CODE: BCSAI405

AI AND INTELLIGENT AGENTS

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following questions. **(10 Marks)**

Q. 1. Answer the following questions: **(1 × 5 = 5 Marks)**

(i) What is the primary goal of Artificial Intelligence?

(ii) Name one statistical learning method besides Naïve Bayes.

(iii) Provide one example of a swarm intelligent system application.

(iv) Name two challenges faced during expert system development.

(v) The A* algorithm combines ____ and ____ costs to find the optimal path.

Q. 2. Answer the following questions: **(1 × 5 = 5 Marks)**

(i) Name two industries where expert systems are widely used.

(ii) What is an expert system?

(iii) Name one application of fuzzy logic in real-world AI systems.

(iv) What does HMM stand for, and where is it used?

(v) State True or False for 'Naïve Bayes models assume independence between features'.

SECTION – B

NOTE:Attempt all questions. **(5×2×2 = 20 Marks)**

Q.3.(i)(a) Compare and contrast uninformed and informed search strategies.

OR

(b) Describe Hidden Markov Models and their application areas.

(ii)(a) Explain the concepts of state space, states, and operators in AI problem solving.

OR

(b) Discuss alpha-beta pruning in game-playing algorithms. How does it improve efficiency?

Q.4.(i)(a) What challenges are faced during the development of Expert Systems?

OR

(b) Describe the role of heuristic functions in search strategies with examples.

(ii)(a) What is the Best-First Search algorithm? How does it differ from A*?

OR

(b) Compare supervised, unsupervised, and inductive learning methods.

SECTION – C

NOTE: Attempt all questions. **(10 × 2 = 20 Marks)**

Q.5.(a) What is Bayes' Rule? Illustrate its application in AI with an example.

OR

(b) Explain how do fuzzy sets and fuzzy logic contribute to AI systems?

Q.6.(a) Outline the stages involved in developing an Expert System, discussing the challenges faced during each stage. Then, explain the concept of swarm intelligence and provide examples of how it is applied in solving complex real-world problems.

OR

(b) Compare and contrast the A* algorithm and Iterative Deepening A* (IDA*). Explain their working principles, advantages, and limitations.